ProbOS — Mission, Vision, and Industry Roadmap

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A note on how to read this document

This document states ProbOS's mission and long-term vision, then sequences which industries it will serve and in what order, honestly distinguishing three categories throughout:

- **Validated** – backed by actual completed work (a literature-validated model, a passing test, a measured number), not just plausibility.
- **Structurally ready** – the underlying kernel already supports this with little or no new engineering, but the domain-specific model itself has not been built or validated yet.
- **Aspirational** – a real, considered target, but with no work done toward it yet and no practitioner conversation confirming the need.

No claim in this document is marked Validated unless it corresponds to something that actually exists and has been tested, per `docs/standards/quality_standards.md`.

Mission

ProbOS treats uncertainty as a first-class data type in software that models safety-critical physical systems. Where deterministic simulation uses one parameter value where there should be a distribution, ProbOS propagates the full range of physically plausible values through a model and reports the tail risk that a single-point simulation cannot see.

Vision

A general-purpose, open, auditable probabilistic execution runtime that becomes the default tool for engineers who must not only simulate a safety-critical system, but *prove to a regulator* that they understand its failure modes – with a causal record from every dangerous predicted outcome back to the specific parameters and assumptions that produced it.

The core thesis: why safety-critical UQ, specifically

ProbOS does not attempt to be a general replacement for existing probabilistic programming frameworks (Stan, PyMC, NumPyro, Pyro, Turing.jl), which are mature, widely used, and well-suited to statistical inference and machine learning workflows. ProbOS's distinct positioning rests on four criteria, present together in a specific, underserved niche:

1. A formal regulator requires documented failure-mode analysis.
2. The system is modelable as an ODE/SDE with genuine, quantifiable parameter uncertainty (material properties, manufacturing variance, patient variability).
3. Failure consequences are rare but catastrophic – exactly the tail risk a deterministic point-estimate simulation is structurally blind to.
4. Existing deterministic tools in that industry are documented to miss these tail risks.

This is a narrower, more specific claim than "uncertainty quantification is useful" – it identifies exactly which industries have this combination of properties, and treats industries lacking one or more of them as a poor fit, not a target to chase for breadth.

Validated today

Battery safety (EV, grid storage) – Validated

`BatteryModel2Cell`, an 8-state Arrhenius thermal-runaway ODE with 15 uncertain parameters, validated against Kim et al. (2007) accelerating rate calorimetry data (Month 1 Week 2). Sobol sensitivity analysis (Month 1 Week 3) identifies `Ea_SEI` as the dominant driver of thermal-runaway variance ($S_1 = 0.457$), and Monte Carlo propagation shows the P95 worst-case cell decomposes 11.6x faster than the P50 median – a tail risk invisible to a deterministic simulation using literature-mean parameters. The provenance tracker (Month 1 Week 3) traces any dangerous predicted output back to the specific parameter draws that produced it, directly serving the "documented failure-mode analysis" requirement regulators impose.

Structurally ready – near-zero additional engineering

Pharmaceutical stability / shelf-life (FDA CMC, ICH Q1A) – Validated

`PharmaStabilityModel` (Month 3 Week 11) uses Avrami kinetics (a related but distinct reaction-kinetics law from `BatteryModel2Cell`'s first-order depletion, since the real, best-validated dataset found required it), validated against Gonzalez-Gonzalez et al. (*Molecules* 2023, 28(23), 7925), a real, peer-reviewed, ICH-referenced chlorhexidine stability study. The activation energy (Ea = 18.52 kcal/mol) is independently cited and cross-checked against an independent 1987 PhD thesis range (16.5–22.9 kcal/mol). The model reproduces the paper's 3 real reported 365-day degradation percentages (5C/25C/30C) to within 1.18 percentage points.

Honest methodological caveat: unlike Kim (2007)'s true forward replication for the battery model, this validation fits the pre-exponential factor A against the paper's own reported summary results, with the Avrami exponent n fixed at 1.0 as an explicit simplifying assumption (confirmed necessary: an unconstrained fit of both A and n against only 3 single-timepoint data points is fundamentally underdetermined, and converged to a physically absurd result when attempted directly). This is a real but weaker validation standard than Kim (2007) and should not be presented as equivalent rigor.

Sobol sensitivity analysis (using the paper's own reported Ea standard error, ±2.61 kcal/mol, as a genuine literature-backed uncertainty) identifies Ea as overwhelmingly dominant ($S_1 = 0.993$), and Monte Carlo propagation reveals a real tail risk invisible to deterministic simulation: the worst-case 5% of outcomes show complete potency loss within one year, versus 82.6% potency remaining at the deterministic point estimate.

Automotive functional safety / EV battery management (ISO 26262) – Structurally ready

This is not a new model – it is the existing, already-validated battery model, reframed for a different regulator. ISO 26262's hazard-analysis and documented-failure-mode requirements are structurally analogous to what FDA/NRC already require, and `BatteryModel2Cell` already satisfies the underlying technical need. The primary remaining work is positioning and outreach specific to automotive/EV safety engineers, not new engineering.

Aspirational – real targets, no work done yet

Medical device (implantable battery safety, FDA 510k) – Aspirational

Named in `docs/sbir_phase1_onepager.md` as an SBIR Phase I target (FDA Q-Sub filing for implantable battery MRI safety). No domain-specific model has been built or validated for this industry yet; the provenance tracker's audit-trail capability is directly relevant once a specific device model exists.

Nuclear safety analysis (NRC) – Aspirational

Named as a target market throughout project documentation since Month 1. No nuclear-specific model exists. NRC's uncertainty quantification requirements match the core thesis structurally, but require a new physical model (reactor thermal/neutronics or similar) not yet attempted.

Aerospace / aviation (FAA, DO-178C / DO-254) – Aspirational

Structurally the closest sibling to the current battery/medical/ nuclear framing: same regulatory pattern (documented, auditable failure analysis), same tail-risk stakes. Would require new domain models (structural fatigue, engine thermal, or flight-control parameter uncertainty) – none built yet.

Grid / power systems reliability (NERC) – Aspirational, higher effort

A real and growing need given renewable-generation variability introducing genuine parameter uncertainty into grid stability models, and NERC's post-incident investigation requirements resembling the provenance/audit-trail pitch already made for medical devices. Rated higher-effort than the above because it requires a new class of stochastic model (demand/generation processes), not an ODE-based physical model like the others.

Explicitly out of current scope

Quantitative finance and robotics are named in earlier project documentation as market categories, but honest reassessment (Month 3) concluded neither is currently a good fit: the finance example model (`OptionPricerModel`, Black-Scholes GBM) is a toy validation case rather than something serving a real quant research audience already well-served by mature tools, and no robotics-specific model exists at all. These are removed from active targeting, not merely deprioritized.

Month-by-month roadmap

Completed (Months 1–3, Weeks 1–10)

- **Month 1 (Weeks 1–4):** Kernel foundation – `Distribution`/`Model` ABCs, `BatteryModel2Cell` validated against Kim (2007), `MonteCarloEngine`, `SobolSensitivity`, `ProvenanceTracker`, C++/OpenMP kernel (7x measured speedup), PDSL v0.1, three cross-discipline forward-simulation examples.
- **Month 2 (Weeks 5–8):** `ParticleFilter` (validated against an exact closed-form Kalman filter), pybind11 bindings, FastAPI service layer, cross-discipline sequential filtering (ED queue, clinical trial), plus a comprehensive mid-month hardening pass closing several real, previously undiscovered gaps (mypy/ruff scope, a license mismatch, vestigial code).
- **Month 3 (Weeks 9–10, in progress):** REST API model registry extended to all four models; GPU kernel path built, honestly benchmarked, and root-cause-optimized via kernel fusion – C++/OpenMP remains the fastest engine for this workload on this hardware, reported truthfully rather than oversold.

Near-term (rest of Month 3 / Month 4) – Structurally ready extensions

- Pharmaceutical stability model, reusing `BatteryModel2Cell`'s Arrhenius architecture with new priors, validated against a real stability-study dataset.
- Automotive/EV safety positioning and outreach, using the existing validated battery model under an ISO 26262 framing.
- PDSL v0.2 (control flow), continuing the priority order already established in Month 3's original plan.

Mid-term (Months 5–6) – Validation, not more building

- At least one real conversation with a practicing battery, pharmaceutical, or automotive safety engineer, confirming (or correcting) the assumed pain point and whether ProbOS's current capabilities actually address it.
- Medical device domain model, informed by whatever is learned from the above conversations rather than built speculatively in advance of them.

Longer-term (Month 7+) – Contingent on validated demand

- Nuclear and aerospace domain models, pursued only if the mid-term validation step confirms genuine practitioner interest in this pattern – not built speculatively for market-size breadth.
- Grid/power systems reliability, the highest-effort item on this list, deferred until the lower-effort extensions have actually been validated.

Honest open questions

This document deliberately does not claim ProbOS is close to displacing established tools, or that its market thesis is confirmed. Two open questions remain genuinely unresolved as of this writing:

1. **Competitive differentiation.** What, specifically, does ProbOS do that Stan, PyMC, or NumPyro do not, for an engineer who already has access to those tools? The provenance/audit-trail angle is the strongest current answer, but has not been tested against a real user's actual workflow.

2. **Practitioner validation.** Every domain claim in this document, including the Validated battery-safety work, has been checked against literature data, not against a live conversation with someone who would actually adopt the tool. This is the single highest-priority open item ahead of further domain expansion.